

RKDF University
Ph.D. Entrance Examination Syllabus
Zoology

Ecology

Characteristics –Natality, Mortality, Density, and age distribution, population control, life-tables, Food chains, Food webs and Ecological pyramids. Air pollution, Water pollution and Oil pollution. Noise pollution and Thermal pollution, Pond as a Ecosystem, energy flow and ecological succession. Habitats – Terrestrial – Aquatic – Marine, Fresh water and estuary, Environmental resources- renewable and non renewable resources. Forest resources- Protection – Chipko movement- A forestation. Wild life management- Wild life sanctuaries and National Parks.

Endocrinology:

General Principles of Endocrinology What are hormones, types of release, Techniques for Studying Endocrinology, Thyroid Hormones

Animal physiology & developmental biology and immunology

Nutrition –types, Enzymes – Enzyme action, Coenzymes, Digestion in man. Respiration – Respiratory pigments, role in transport of O₂ and CO₂ in man. Circulation - blood composition, origin and conduction of heart beat in an – blood pressure, Heart diseases– heart attack.

Cell & Molecular biology

Prokaryotic and eukaryotic cells –Ultrastructure and Organization. Plasma membrane–Ultra structure–Chemical composition and functions of modifications of plasma membrane. Endoplasmic reticulum: Morphology, Ultra structure, chemical composition and functions. Golgi complex: Ultra structure, chemical composition and functions.

mitosis, meiosis and cell cycle. Cell - Chemical nature and macromolecular protein structure and function; membrane architecture, membrane associated process, ATP synthesis and photosynthesis, Sub-cellular organelles - mitochondria, chloroplast, DNA replication - prokaryotic and eukaryotic DNA replication - Mechanisms of

DNA replication - Enzymes and accessory proteins involved in DNA replication - Bacterial genetic system - transformation, conjugation and transduction, Transcription & Translation

Immunology & recombinant DNA technology

Historical perspectives - overview of immune system, innate and acquired immunity, immune - systematic structure and organization.

Antigen and antigenicity, Immunoglobulins - structure, complements, antigen - antibody interaction - monoclonal antibodies.

History of Recombinant DNA Technology. Isolation and Quantification of DNA and RNA. Host controlled Restriction –Modification system, Restriction Endonucleases. Cutting and joining of DNA molecules *in vitro*. Phosphatases, Ligases and Polymerases. Vectors: Plasmid, Bacteriophage, Cosmids, Phagemid and other (SV 40) Virus vectors. Expression Vectors.

INVERTEBRATA & CHORDATA

Phylum Porifera: General characters- Type study- Ascon- Cellular structure, Phylum Platyhelminthes: General Characters- Classification- Type study- Liver fluke- Structure and Reproduction, Type Study: Amphioxus- external characters, digestive, excretory, respiratory, and circulatory systems, Class : Amphibia :General characters and classification -Type Study : Frog –External characters

MICROBIOLOGY AND BIOCHEMISTRY

Classification of microorganisms- General characteristics of Bacteria, Virus, Yeast. Bacteria- Morphology, Bacterial cell structure, Motility, Nutrition and Reproduction. Virus-discovery- Morphology, Classification, phages and life cycle. Yeast-Morphology, cell structure, Multiplication, phages and cycle.

Genetics

DNA as genetic material, Structure and types of DNA and RNA, Genetic code, Protein synthesis
Transcription and translation

DNA Replication and Gene Structure

DNA replication, Cis-trans complementation test, Fine structure analysis of r II region of T4 by Benzer.

Mutation

Evidence for spontaneous nature of mutation, Molecular basis of mutation- Types of mutation, Types of bacterial mutants and their isolation, Mutagenic agents- Physical and chemical, Mutation rate and Ames test.